

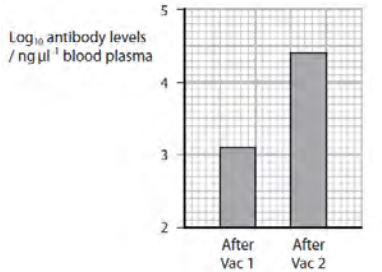
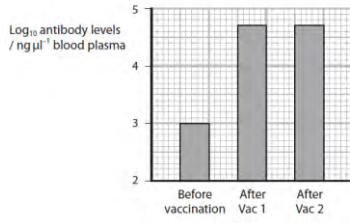
Question number	Answer	Additional guidance	Mark																			
8(a)	<table><tr><th rowspan="2">Statement</th><th colspan="4">Type of artificial immunity</th></tr><tr><th>both active and passive</th><th>active only</th><th>passive only</th><th>neither active nor passive</th></tr><tr><td>Antigens are injected into the person</td><td>☐</td><td>☒</td><td>☐</td><td>☐</td></tr><tr><td>Immunity is long term</td><td>☐</td><td>☒</td><td>☐</td><td>☐</td></tr></table>	Statement	Type of artificial immunity				both active and passive	active only	passive only	neither active nor passive	Antigens are injected into the person	☐	☒	☐	☐	Immunity is long term	☐	☒	☐	☐		(2)
Statement	Type of artificial immunity																					
	both active and passive	active only	passive only	neither active nor passive																		
Antigens are injected into the person	☐	☒	☐	☐																		
Immunity is long term	☐	☒	☐	☐																		

Question number	Answer	Additional guidance	Mark
8(b)	A description that includes two of the following points: - T (helper) cells release cytokines (to stimulate B cells) - OR - T (helper) cells activate B cells (1) - B cells {differentiate into / specialise into} plasma cells (that produce antibodies) (1)	ACCEPT CD4 cells DO NOT ACCEPT cytokinins ACCEPT description e.g. causing B cells to divide IGNORE T killer cells activated DO NOT ACCEPT {make / produce / divide into / clonal expansion into} plasma cells	(2)

Question number	Answer	Additional guidance	Mark
8(c)(i)	An explanation that includes two of the following points: • both groups received same dose of vaccine so that there was the same number of antigens (1) • nobody should have a disorder of the immune system otherwise an immune response may be affected (as effectively) (1) • {levels of antibodies measured / vaccines given} at same time intervals as antibody levels change (with time) (1) • people in group 2 should all have had the infection at a similar time otherwise the antibody levels maybe different (1)	ACCEPT volume / concentration / mass affect the immune system to the same degree ACCEPT other reasons with the explanation : do not have {HIV / weakened immune system} appropriate named variable relating to people in group e.g. age, gender, obesity people are not taking {drugs / medication} people do not have an infection	(2)

Question number	Answer	Additional guidance	Mark
8(c)(ii)	1258.925412 and 25118.8643 (1) 19.95 / 20 (x) (1) OR (4.4 - 3.1 =) 1.3 (1) 19.95 / 20 (x) (1)	Bald answer of 19.95 / 20 (x) = 2 marks Bald answer of 1.3 = 1 mark Bald answer of 23859.9389 rounded correctly = 1 mark Bald answer of 19.9526 = 1 mark	(2)

Question number	Answer	Additional guidance	Mark
8(c)(iii)	there will not be any antibodies (in the blood plasma) if the person had not been previously infected / if they had not been infected the immune system would not have been stimulated (1)	ACCEPT no plasma cells that produce this antibody DO NOT ACCEPT antibodies to kill virus antibiotics	(1)

Question number	Answer														
c)(iv)	Indicative content: Group 1 Vac 1 • antibody levels increase following vac 1 because (primary) immune response initiated • credit details of primary immune response • resulting in plasma cells releasing antibodies Vac 2 • vac 2 contains antigens that were in vac 1 • stimulate secondary immune response • because memory cells present • therefore results in higher levels of antibody released • could be other antigens in vac 2 which stimulated production of other types of antibodies • so both stimulated a primary immune response Group 2 Before vaccination • antibodies present as person previous infected so primary immune response generated Vac 1 • vac 1 stimulated secondary immune response • memory cells formed from infection • so levels of antibodies produced were higher • could have been other antigens present in vac 1 • contributing to higher levels of antibody Vac 2 • vac 2 resulted in same levels of antibodies suggesting that antigens in vac 1 and vac 2 were the same • but the time lapse between vaccination and testing was long enough for antibody levels to decrease • vac 2 may have had different antigens to vac 1 but still stimulated secondary response  Log₁₀ antibody levels / ng µl⁻¹ blood plasma <table border="1" data-bbox="1615 520 1995 568"> <thead> <tr> <th>Vaccination</th> <th>Log₁₀ antibody levels</th> </tr> </thead> <tbody> <tr> <td>After Vac 1</td> <td>3.1</td> </tr> <tr> <td>After Vac 2</td> <td>4.4</td> </tr> </tbody> </table> Group 1: People who had not been infected before the vaccine was given  Log₁₀ antibody levels / ng µl⁻¹ blood plasma <table border="1" data-bbox="1715 1015 2063 1062"> <thead> <tr> <th>Vaccination</th> <th>Log₁₀ antibody levels</th> </tr> </thead> <tbody> <tr> <td>Before vaccination</td> <td>3.0</td> </tr> <tr> <td>After Vac 1</td> <td>4.7</td> </tr> <tr> <td>After Vac 2</td> <td>4.7</td> </tr> </tbody> </table> Group 2: People who had been infected with the virus before the vaccine was given	Vaccination	Log ₁₀ antibody levels	After Vac 1	3.1	After Vac 2	4.4	Vaccination	Log ₁₀ antibody levels	Before vaccination	3.0	After Vac 1	4.7	After Vac 2	4.7
Vaccination	Log ₁₀ antibody levels														
After Vac 1	3.1														
After Vac 2	4.4														
Vaccination	Log ₁₀ antibody levels														
Before vaccination	3.0														
After Vac 1	4.7														
After Vac 2	4.7														

			Additional guidance
evel 0	0	No awardable content	
evel 1	1-2	An explanation may be attempted but with limited interpretation or analysis of the scientific information and with a focus on mainly just one piece of scientific information. The explanation will contain basic information, with some attempt made to link knowledge and understanding to the given context.	Simple explanation of data 1 mark = some description of the data in one graph 2 marks = some description of the data in both graphs
evel 2	3-4	An explanation will be given, with occasional evidence of analysis, interpretation and/or evaluation of both pieces of scientific information. The explanation shows some linkages and lines of scientific reasoning, with some structure.	Some explanation given using the information given / own knowledge 3 marks = simple explanation of either primary or secondary immune response OR of either graph 1 or graph 2 4 marks = simple explanation of both primary and secondary immune response OR of both graph 1 and graph 2 OR detailed explanation of one aspect (only)
evel 3	5-6	An explanation is made that is supported throughout by sustained application of relevant evidence of analysis, interpretation and/or evaluation of both pieces of scientific information. The explanation shows a well-developed and sustained line of scientific reasoning, which is clear and logically structured.	Data logically explained with extended use of information provided 5 marks = detailed and accurate explanation of one graph and simple explanation of the other graph 6 marks = clear, detailed and accurate explanation of both graphs

